

MAT205: Abstract Algebra II

Conjugate Elements and the Isomorphism Extension Theorem

Ma, Jia-Jun - Xiamen University Malaysia

Part I

Conjugates, Extensions, and the Index

Motivation: Lifting Isomorphisms Along Algebraic Extensions

L12b: Universal Property of Algebraic Closure gave: every algebraic extension E/F embeds into $\overline{F}$ via some $\sigma : E \hookrightarrow \overline{F}$ with $\sigma|_F = \text{id}_F$.

Question. What if F itself carries an automorphism, or more generally an isomorphism $F \rightarrow F'$? Can we lift it to an embedding $E \rightarrow \overline{F}'$?

Answer (today). Yes – the **Isomorphism Extension Theorem**.

Two consequences we will derive:

- the full uniqueness statement for algebraic closures (L12b only handled $\sigma = \text{id}_F$);
- a method to count F -automorphisms by counting choices of conjugate roots – the starting point of Galois theory.

Definition: Conjugate Elements

Recall (L12). For α algebraic over F , the **minimal polynomial** $m_\alpha \in F[x]$ is the unique monic polynomial of least degree with $m_\alpha(\alpha) = 0$. Equivalently, m_α generates the ideal $\{f \in F[x] : f(\alpha) = 0\}$ in the PID $F[x]$. It is **irreducible**.

Let E/F be a field extension and $\alpha, \beta \in E$ both algebraic over F .

Definition. α and β are **conjugate over F** if their minimal polynomials agree: $m_\alpha = m_\beta$ in $F[x]$.

Equivalently: α and β are both roots of the same monic irreducible polynomial $p \in F[x]$ (necessarily $p = m_\alpha = m_\beta$).

Examples.

- $\sqrt{2}$ and $-\sqrt{2}$ over $\mathbb{Q}$ (roots of $x^2 - 2$).
- i and $-i$ over $\mathbb{R}$ (roots of $x^2 + 1$).
- $\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}$ with $\omega = e^{2\pi i/3}$ over $\mathbb{Q}$ (roots of $x^3 - 2$).
- $a + bi$ and $a - bi$ over $\mathbb{R}$ for $a, b \in \mathbb{R}, b \neq 0$ (classical complex conjugates).

Lemma: Minimal Polynomial Under Base Change

Setup. $F \subseteq M$ a tower of fields, α algebraic over F (in some extension of M).

Lemma. α is algebraic over M , and

$$[M(\alpha) : M] \leq [F(\alpha) : F].$$

Proof.

- $m_{\alpha,F} \in F[x] \subseteq M[x]$ vanishes at α , so α is algebraic over M with minimal polynomial $m_{\alpha,M} \in M[x]$.
- $m_{\alpha,M}$ generates the ideal $\{g \in M[x] : g(\alpha) = 0\}$ in the PID $M[x]$; since $m_{\alpha,F}$ is in this ideal, $m_{\alpha,M} \mid m_{\alpha,F}$ in $M[x]$.
- $\deg m_{\alpha,M} \leq \deg m_{\alpha,F}$.
- L12 (Minimal Polynomial recall): $[F(\alpha) : F] = \deg m_{\alpha,F}$, $[M(\alpha) : M] = \deg m_{\alpha,M}$. Combine. $\square$

Equality iff $m_{\alpha,F}$ remains irreducible in $M[x]$.

Examples.

Theorem: Conjugation Isomorphism

Conventions. $\varphi : E_1 \rightarrow E_2$ is an **F -homomorphism** if φ is a ring homomorphism with $\varphi|_F = \text{id}_F$. **F -isomorphism** = bijective, **F -embedding** = injective.

Theorem (Fraleigh 48.3). Let α, β be algebraic over F and conjugate over F . There exists a unique F -isomorphism

$$\psi_{\alpha,\beta} : F(\alpha) \rightarrow F(\beta), \quad \psi_{\alpha,\beta}(\alpha) = \beta.$$

Explicit formula. Every element of $F(\alpha)$ has the form $c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1}$ where $n = \deg m_\alpha$ (L12: Minimal Polynomial and $F(\alpha)$), and

$$\psi_{\alpha,\beta}(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1}) = c_0 + c_1\beta + \cdots + c_{n-1}\beta^{n-1}.$$

Proof. L12: Minimal Polynomial and $F(\alpha)$ gives canonical F -isomorphisms $\varphi_\alpha : F[x]/(m_\alpha) \xrightarrow{\sim} F(\alpha)$ and $\varphi_\beta : F[x]/(m_\beta) \xrightarrow{\sim} F(\beta)$ via evaluation. Conjugacy gives $m_\alpha = m_\beta$, so the source quotient rings are equal. Set $\psi_{\alpha,\beta} := \varphi_\beta \circ \varphi_\alpha^{-1}$.

Theorem: Isomorphism Extension

Setup. $\sigma : F \rightarrow F'$ an isomorphism of fields. $\overline{F'}$ an algebraic closure of F' . E/F an algebraic extension.

Theorem (Fraleigh 49.3). σ extends to a field embedding $\tau : E \hookrightarrow \overline{F'}$ with $\tau|_F = \sigma$.

Slogan. Any isomorphism between base fields lifts to an embedding of an algebraic extension into any chosen algebraic closure of the codomain.

Specialization to L12b: Universal Property of Algebraic Closure. $F = F', \sigma = \text{id}_F, \overline{F'} = \overline{F}$: recovers the universal property of $\overline{F}$.

Specialization to the Conjugation Isomorphism above. $E = F(\alpha), F' = F, \sigma = \text{id}_F, \overline{F'} = \overline{F}$: gives the conjugation $\psi_{\alpha,\beta}$ composed with $F(\beta) \hookrightarrow \overline{F}$.

Proof Skeleton: The Zorn Argument

Step 1 (Poset). $\mathcal{P} = \{ (K, \tau_K) : F \subseteq K \subseteq E, \tau_K : K \rightarrow \overline{F'}$ embedding, $\tau_K|_F = \sigma \}$, ordered by extension of pairs.

Step 2 (Chains have upper bounds). For a chain $\{(K_i, \tau_i)\}$, set $K_\infty = \bigcup K_i$ and $\tau_\infty(x) = \tau_i(x)$ for $x \in K_i$ (well-defined by chain compatibility). $(K_\infty, \tau_\infty) \in \mathcal{P}$.

Step 3 (Apply Zorn). Get a maximal $(K_{\max}, \tau_{\max}) \in \mathcal{P}$.

Step 4 (Maximal forces $K_{\max} = E$). If $K_{\max} \subsetneq E$, pick $\alpha \in E \setminus K_{\max}$, algebraic over $K_{\max}$ with minimal polynomial $m \in K_{\max}[x]$. Let $m^{\tau_{\max}} \in \overline{F'}[x]$ be the polynomial with coefficients $\tau_{\max}$ -imaged. It has a root $\beta \in \overline{F'}$ (algebraically closed).

Extend $\tau_{\max}$ to $\tau' : K_{\max}(\alpha) \rightarrow \overline{F'}$ by $\tau'(\alpha) = \beta$. Then $(K_{\max}(\alpha), \tau') \in \mathcal{P}$ strictly extends $(K_{\max}, \tau_{\max})$ – contradicts maximality.

Conclusion. $K_{\max} = E$, and $\tau_{\max} : E \rightarrow \overline{F'}$ is the required extension. $\square$

Corollary: Generalized Uniqueness of Algebraic Closure

Corollary 1 (Lifting conjugations; Fraleigh 49.4). Let E/F be algebraic and $\alpha, \beta \in \overline{F}$ conjugate over F . The conjugation isomorphism $\psi_{\alpha, \beta} : F(\alpha) \rightarrow F(\beta)$ extends to an embedding $E \rightarrow \overline{F}$.

Corollary 2 (Uniqueness, full version; Fraleigh 49.5). Let $\sigma : F \rightarrow F'$ be an isomorphism. Then σ extends to an isomorphism $\overline{F} \xrightarrow{\sim} \overline{F'}$. Taking $\sigma = \text{id}_F$ recovers the L12b: Universal Property of Algebraic Closure corollary; for general σ , isomorphic base fields have isomorphic algebraic closures.

Application: $|\text{Aut}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})| = 4$.

Any $\mathbb{Q}$ -automorphism σ sends $\sqrt{2}$ to a root of $x^2 - 2$, so $\sigma(\sqrt{2}) \in \{\pm\sqrt{2}\}$; similarly $\sigma(\sqrt{3}) \in \{\pm\sqrt{3}\}$. Each of the 4 assignments extends to an automorphism by Corollary 1 (twice).

Hence $|\text{Aut}(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})| = 4 = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$. This equality between automorphism count and degree is the starting point of Galois theory.

Theorem: Degree Bound on Embeddings and Automorphisms

Let E/F be a finite algebraic extension, $\overline{F}$ an algebraic closure.

Theorem. $\#\{F\text{-embeddings } E \rightarrow \overline{F}\} \leq [E : F]$.

Corollary. $|\text{Aut}(E/F)| \leq [E : F]$, since every automorphism is an embedding into $\overline{F}$.

Proof (induction on $[E : F]$). Base $[E : F] = 1$ trivial. Inductive step: pick $\alpha \in E \setminus F$, $K = F(\alpha)$, degree $n = [K : F] \geq 2$, $[E : K] < [E : F]$. Each τ restricts to $\tau|_K$, giving at most n restrictions (one per root of m_α in $\overline{F}$, L11: Root Bound over a Field). Each restriction extends to at most $[E : K]$ embeddings (induction hypothesis + isomorphism extension). Multiply: $n \cdot [E : K] = [E : F]$ by tower law (L12: Tower Law). $\square$

Example (sharp). $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$: $|\text{Aut}| = 4 = \#\{\text{embeddings}\} = [E : F]$.

Example (asymmetric). $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$: $\#\{\text{embeddings into } \overline{\mathbb{Q}}\} = 3$ (one per root of $x^3 - 2$), but $|\text{Aut}| = 1$ — $\omega\sqrt[3]{2}, \omega^2\sqrt[3]{2} \notin \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$.

Definition: Index of a Field Extension and Multiplicativity

Definition. For a finite algebraic extension E/F , the **index** is

$$\{E : F\} := \#\{F\text{-embeddings } E \rightarrow \overline{F}\}.$$

The count is independent of the algebraic closure (Fraleigh 49.7).

In this notation, $|\text{Aut}(E/F)| \leq \{E : F\} \leq [E : F]$.

Theorem (Multiplicativity). For $F \subseteq K \subseteq E$ finite algebraic,

$$\{E : F\} = \{E : K\} \cdot \{K : F\}.$$

Proof. Each F -embedding $\tau : E \rightarrow \overline{F}$ restricts to $\tau|_K$, ranging over the $\{K : F\}$ many F -embeddings $K \rightarrow \overline{F}$. For each fixed restriction σ , the iso-extension theorem (Isomorphism Extension above) plus intrinsic invariance (49.7) gives exactly $\{E : K\}$ extensions of σ to E . Sum: $\{E : F\} = \{K : F\} \cdot \{E : K\}$. $\square$

Separable and Inseparable Extensions

Definition. A nonzero $p \in F[x]$ is **separable** if it has no repeated roots in $\overline{F}$, equivalently $\gcd(p, p') = 1$.

Definition. α algebraic over F is **separable over F** if m_α is separable. E/F algebraic is **separable** if every $\alpha \in E$ is separable over F .

Theorem. For finite algebraic E/F :

$$\{E : F\} = [E : F] \iff E/F \text{ is separable.}$$

Char 0 $\Rightarrow$ separable. Let $p \in F[x]$ irreducible, $\text{char } F = 0$. Then $\deg p' = \deg p - 1 \geq 0$ and $p' \neq 0$. Since p is irreducible and $\deg \gcd(p, p') < \deg p$, we get $\gcd(p, p') = 1$. So every irreducible in char 0 is separable; every algebraic extension in char 0 is separable.

Inseparable example (char p). $F = \mathbb{F}_p(t)$, $\alpha = t^{1/p} \in \overline{F}$, $m_\alpha = x^p - t$.

- In $\overline{F}[x]$: $m_\alpha = (x - \alpha)^p$ by Frobenius binomial – single root with multiplicity p .
- $m'_\alpha = px^{p-1} = 0$ in char p , so $\gcd(m_\alpha, m'_\alpha) = m_\alpha \neq 1$: **inseparable**.
- Index: $\{F(\alpha) : F\} = 1$ (only one root to send α to), but $[F(\alpha) : F] = p$ – strict inequality $\{E : F\} < [E : F]$.